

SHUZHAO XIE

Ph.D. Candidate, Tsinghua University | Visiting Research Assistant, Carnegie Mellon University

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🎓 EDUCATION

Tsinghua University

- Ph.D. in Computer Science and Technology. Advisor: Prof. Zhi Wang.

Shenzhen, China

Aug. 2024 – Present

Tsinghua University

- M.Eng. in Computer Technology. Advisor: Prof. Zhi Wang.

Shenzhen, China

Jun. 2020 – Aug. 2023

Beijing Normal University

- B.S. in Computer Science and Technology.

Beijing, China

Aug. 2016 – Jun. 2020

🖥️ RESEARCH & PROFESSIONAL EXPERIENCE

Carnegie Mellon University, School of Computer Science, USA

Apr. 2026 – Oct. 2026

- Visiting Research Assistant, advised by Prof. Minchen Li; Real2Sim and robot simulation.
- Lead **CoDimRecon**: reconstruct editable rigid, articulated, and deformable scenes from multi-view RGB; use geometric priors and generated meshes to guide scene authoring, with explicit joints for articulated objects.
- Build category-wise agents for centerlines with radii, shells with thickness, and watertight volumes. Use reusable skills to initialize physics, then robot tests to refine geometry and behavior, including plastic paper folding.

ABB Robotic Picking Line, Data Collection and Policy Fine-tuning

Nov. 2025 – Dec. 2025

- Built a dual-camera data collection pipeline with **LeRobot**; fine-tuned $\pi_{0.5}$ on the collected data to pick moving objects on the production line.

Tsinghua University, Efficient 3D Representations

Oct. 2023 – Oct. 2025

- Develop **MesonGS/SizeGS/MesonGS++** with mixed precision, mixed-integer optimization, and CUDA-parallel search; achieve up to $8\times$ faster search and over $34\times$ compression while preserving rendering fidelity.

Tsinghua University, Armol / SkyML: Reinforcement Learning

Jul. 2020 – Aug. 2023

- Develop combinatorial RL for multicloud inference: design image-feature states, provider-subset actions, and accuracy-cost rewards. Train **SAC** with replay and twin Q-networks, compare with **PPO/TD3**, and reduce fees by 67% at comparable accuracy. First-author **INFOCOM 2022 / TMM 2024**; code.

Tencent, Shanghai, China

Jan. 2022 – Jun. 2022

- Model Acceleration Intern; worked with Faster-RCNN, ONNX, OpenVINO, and TensorRT.

📄 SELECTED PUBLICATIONS & PREPRINTS (* equal contribution)

- **CoDimRecon: Agentic Reconstruction of Sim-Ready 3D Scenes with Deformable Curves, Surfaces, and Volumes**
Shuzhao Xie*, Lelin Wang*, Guying Lin, Zhi Wang, Minchen Li. *arXiv:2609.36024*, 2026. 🌐
- **SizeGS: Size-aware Compression of 3D Gaussian Splatting via Mixed Integer Programming**
Shuzhao Xie, Jiahang Liu, Weixiang Zhang, Shijia Ge, Sicheng Pan, Chen Tang, *et al.* *ACM MM*, 2025. *Best Paper Candidate*.
- **MesonGS: Post-training Compression of 3D Gaussians via Efficient Attribute Transformation**
Shuzhao Xie, Weixiang Zhang, Chen Tang, Yunpeng Bai, Rongwei Lu, Shijia Ge, Zhi Wang. *ECCV*, 2024.
- **MesonGS++: Post-training Compression of 3D Gaussian Splatting with Hyperparameter Searching**
Shuzhao Xie, Junchen Ge, Weixiang Zhang, Jiahang Liu, Chen Tang, Yunpeng Bai, *et al.* *arXiv preprint*, 2026.
- **SkyML: A MLaaS Federation Design for Multicloud-based Multimedia Analytics**
Shuzhao Xie, Yuan Xue, Yifei Zhu, Zhi Wang. *IEEE TMM*, 2024.
- **Cost Effective MLaaS Federation: A Combinatorial Reinforcement Learning Approach**
Shuzhao Xie, Yuan Xue, Yifei Zhu, Zhi Wang. *IEEE INFOCOM*, 2022.
- **IGen: Scalable Data Generation for Robot Learning from Open-World Images**
Chenghao Gu, Haolan Kang, Junchao Lin, Jinghe Wang, Duo Wu, Shuzhao Xie, *et al.* *CVPR*, 2026.
- **Splatwizard: A Benchmark Toolkit for 3D Gaussian Splatting Compression**
Xiang Liu, Yimin Zhou, Jinxiang Wang, Yujun Huang, Shuzhao Xie, Shiyu Qin, *et al.* *CVPR Findings*, 2026.
- **GeniWorld: A Generalizable Interactive World Model for Robotic Manipulation via Visual Actions**
Chenghao Gu, Hanyang Yu, Jingbo Zhang, Haitao Lin, Wenya Zhang, Jinghe Wang, Hanglei Jin, Shuzhao Xie, Jingyan Jiang, Zhi Wang. *arXiv:2608.06332*, 2026.
- **RoboStream: Weaving Spatio-Temporal Reasoning with Memory in Vision-Language Models for Robotics**
Yuzhi Huang*, Jie Wu*, Weijue Bu*, Ziyi Xiong, Gaoyang Jiang, Ye Li, Kangye Ji, Shuzhao Xie, Yue Huang, Chenglei Wu, Jingyan Jiang, Zhi Wang. *ECCV*, 2026.

🏆 AWARDS & HONORS

- Best Paper Candidate, ACM International Conference on Multimedia (**top 0.5%**). 2025
- Bronze Medal ($\times 2$), ACM ICPC/CCPC Regional Contest. 2018, 2019

🔧 SKILLS

- **Real2Sim / RL**: Multi-view reconstruction, deformable simulation, SAC, reward design, PPO/TD3 evaluation.
- **Engineering**: Python, C/C++, CUDA, Bash, PyTorch; LeRobot, Blender, UIPC, Genesis.